Exam Date & Time: 23-Feb-2026 (01:15 PM - 03:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computer Science and Applications
University Theory Examination (Supplementary Exam-1) February-2026
1st Semester of Bachelor of Computer Applications (BCA)

FOUNDATION OF PROGRAMMING [CAUC101]

Marks: 50 **Duration: 135 mins.**

SECTION-I

Answer all the questions. Section Duration: 20 mins

1 What is an algorithm?

1) A programming language 2) A step-by-step procedure to solve a problem 3) A flowchart symbol 4) A compiler (1)

Course Outcome: "CO1" / Cognitive Level: "Remember"

2 Which symbol is used to include header files in C?

1) < > 2) { } 3) ( ) 4) " " (1)

Course Outcome: "CO2" / Cognitive Level: "Understand"

3 Which function is used to display output in C?

1) scanf() 2) printf() 3) output() 4) write() (1)

Course Outcome: "CO2" / Cognitive Level: "Application"

4 _____ is an example of a loop.

1) for 2) if 3) switch 4) else (1)

Course Outcome: "CO3" / Cognitive Level: "Application"

5 _____ is the symbol of assignment operator.

1) = 2) ? 3) > 4) & (1)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

6 Who developed C language?

1) Dennis Ritchie 2) Adam Williams 3) J. Tenanbaum 4) Steve Jobs (1)

Course Outcome: "CO2" / Cognitive Level: "Remember"

7 _____ symbol is used to represent pointer.

1) * 2) <</td> 3) > 4) ? (1)

Course Outcome: "CO5" / Cognitive Level: "Remember"

8 C follows top-down approach. (1)

1)	True		2)	False	

Course Outcome: "CO2" / Cognitive Level: "Remember"

9 _____ function is used to calculate length of a string.

1)	strcpy()		2)	strcat()		3)	strlen()		4)	strstr()	

(1)

Course Outcome: "CO4" / Cognitive Level: "Understand"

10 Which symbol is used for condition in a flowchart?

1)	diamond		2)	rectangle		3)	circle		4)	arrow	

(1)

Course Outcome: "CO1" / Cognitive Level: "Remember"

SECTION-II

Answer 4 out of 7 questions.

1 Explain the Sequence, Decision, and Repetition features ofalgorithms with examples.

(5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

2 List any four symbols used in a flowchart and explain theirpurpose.

(5)

Course Outcome: "CO2" / Cognitive Level: "Remember"

3 Write the algorithm to check whether a number is odd or even.

(5)

Course Outcome: "CO1" / Cognitive Level: "Create"

4 Draw a structure of a C program and evaluate each section(Documentation, Link, Definition, Global Declaration, main(),Subprograms).

(5)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5 Explain different operators in C.

(5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

6 Write the syntax of a simple if statement and explain when it isused.

(5)

Course Outcome: "CO3" / Cognitive Level: "Application"

7 Provide syntax and explain while, do...while and for loop.

(5)

Course Outcome: "CO3" / Cognitive Level: "Create"

SECTION-III

Answer 4 out of 7 questions.

1 Explain one-dimensional arrays with declaration, initialization(compile time & run time), and examples. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

2			Explain any five string functions (strcat, strncat, strcmp, strcpy,strlen, etc.) with syntax, purpose, and examples. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

3			Discuss the three major elements of a user-defined function:function declaration, function call, and function definition, withexamples. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

4			Describe the four categories of functions and provide suitableexamples. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

5			Define a pointer. How is a pointer variable declared in C? (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

6			Give difference between arrays and structures. (5)

Course Outcome: "CO4" / Cognitive Level: "Remember"

7			Write a C program code to make addition, subtraction, division andmultiplication of two numbers in individual user defined function. (5)

Course Outcome: "CO1" / Cognitive Level: "Create"

-----End-----

Event Name	Subject
University Exam Dec-Feb 2026	CAUC101:FOUNDATION OF PROGRAMMING

#	Course Outcome	Description	Marks
1	CO1	Be able to solve problems using algorithms and flowcharts.	17
2	CO2	Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants.	14
3	CO3	Gain the knowledge of different conditional and iterative statements and expressions.	16
4	CO4	Explore the utilization and manipulation of numbers and characters arrays both.	17
5	CO5	Be able to understand user define function. Also, learn Programing-using Array with Pointer and Structure.	16
		Total	**80**

* NA - CO Not Selected

Event Name	Subject
University Exam Dec-Feb 2026	CAUC101:FOUNDATION OF PROGRAMMING

#	Level	No of Questions	Marks (%)
1	Remember	7	15(18.75%)
2	Understand	4	12(15%)
3	Application	4	12(15%)
4	Analysis	2	10(12.5%)
5	Create	3	15(18.75%)
6	Evaluate	4	16(20%)
	Total	24	80 (100%)